

Identification Options for a Scalar Triangular System

A self-contained guide using the notation of the Lewbel-type set-identification note

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Abstract

This report translates the identification ideas in Lee, Lewbel, Schennach, and Zhang’s `trigmm` paper and Daniel Lewis’s Annual Review survey on higher-moment identification into the scalar triangular setting used in the Lewbel-type note supplied with the prompt. The maintained empirical object is a linear triangular model with one outcome, one endogenous regressor, one scalar external or constructed instrument, and a common set of controls. I separate three logically different uses of higher moments: (i) Lewbel’s heteroskedasticity-generated instrument, which is the baseline for the notation; (ii) the Lee–Lewbel–Schennach–Zhang common-factor method, which can point identify the coefficient by non-Gaussianity or set identify it by second moments; and (iii) the broader SVAR/ICA options surveyed by Lewis, which can be translated to this two-observable system only after imposing a square independent-shock representation. The report emphasizes what is identified, which assumption supplies the missing information, how a single scalar instrument enters, and what form of inference or software support is available.

Contents

1	Executive summary	2
2	Target scalar triangular system and notation	4
3	Baseline: Lewbel’s scalar heteroskedasticity method	5
3.1	Exact validity gives a point estimate	5
3.2	Bounded invalidity gives a scalar interval	6
3.3	Sampling uncertainty	6
4	Direct option 1 from Lee–Lewbel–Schennach–Zhang: point identification by non-Gaussianity	7
4.1	Cumulant notation	7
4.2	Identification from two cumulant equations	8
4.3	How one scalar external instrument can enter	8
4.4	Inference and computation	9
5	Direct option 2 from Lee–Lewbel–Schennach–Zhang: second-moment set identification	9
5.1	Important consequence: the marginal set for θ is usually not a finite interval	10
5.2	Confidence region used by <code>trigmmset</code>	10
5.3	Coefficients on controls	11

6	Options from Lewis’s review, translated to the scalar triangular setting	11
6.1	Heteroskedasticity option A: two variance regimes	11
6.2	Heteroskedasticity option B: more regimes, Markov switching, smooth transition, and GARCH	12
6.3	Heteroskedasticity option C: unconditional moments of squared residuals	12
6.4	Non-Gaussianity option A: independent component analysis	13
6.5	Non-Gaussianity option B: likelihood and pseudo-likelihood	13
6.6	Non-Gaussianity option C: moment-based GMM with coskewness or cokurtosis	13
6.7	Non-independent components and coheteroskedasticity	13
7	Using one scalar instrument with each option	14
8	Inference and weak-identification issues	14
9	Software and implementation notes	15
10	A practical decision tree	15
11	Conclusion	16

1 Executive summary

The target coefficient is the scalar structural effect of the endogenous regressor, denoted by θ . The most useful summary is the following.

Option	Main identifying information	What is identified in the scalar triangular model	Main practical cautions
Lewbel exact heteroskedasticity	A scalar Z satisfies $\text{Cov}(Z, \varepsilon_1 \varepsilon_2) = 0$ and $\text{Cov}(Z, \varepsilon_2^2) \neq 0$.	Point identification: $\theta = \text{Cov}(Z, \omega_1 \omega_2) / \text{Cov}(Z, \omega_2^2)$.	Needs heteroskedastic relevance of the first-stage residual. With one instrument it is just identified, so overidentification tests are unavailable.
Lewbel bounded-invalidity set	Same scalar Z , but $\text{Corr}(Z, \varepsilon_1 \varepsilon_2)$ is allowed to be nonzero, bounded relative to $\text{Corr}(Z, \varepsilon_2^2)$.	A finite interval for θ when the relevance covariance is nonzero and the slack is below one.	The bound depends on a user-chosen slack τ ; sampling uncertainty requires an outer confidence construction or bootstrap.

Option	Main identifying information	What is identified in the scalar triangular model	Main practical cautions
Lee–Lewbel–Schennach–Zhang point identification (<code>trigmm</code>)	Common-factor errors $\omega_2 = u + v$, $\omega_1 = \alpha u + \theta v + r$ with independent components and enough non-Gaussianity.	Point identification of (α, θ) from two higher-cumulant equations $M_p(\alpha, \theta) = 0$; then all other coefficients follow from the common residualization map.	Requires the sign of the common-factor loading and nondegenerate higher cumulants. Nonlinear GMM can have local minima and weak higher-moment identification.
Lee–Lewbel–Schennach–Zhang sharp set	Common-factor structure without the non-Gaussianity point-identifying condition; uses the implied decomposition of observables into Gaussian and non-Gaussian independent parts.	A sharp population joint set for the latent common-factor parameters.	Conceptually important but hard to estimate and interpret; the Stata paper therefore reports a simpler covariance-based outer bound.
Lee–Lewbel–Schennach–Zhang set identification (<code>trigmmset</code>)	Same common-factor structure and sign of common-factor loading, but no non-Gaussianity; only second moments.	A joint set in (α, θ) . For positive common-factor loading, $\theta \leq B_0 \leq \alpha$ and $(B_0 - \theta)(\alpha - B_0) \leq D_0$. The projection onto θ alone is generally one-sided, not finite.	The reported confidence set is conservative and is a joint region; a finite interval for θ alone needs an extra magnitude restriction on the common-factor loading.
Lee–Lewbel–Schennach–Zhang plus an external instrument	Add a conventional IV moment such as $\mathbb{E}[Z(\omega_1 - \theta\omega_2)] = 0$ to the internal non-Gaussian GMM moments.	Usually point identification, often overidentification, with a direct comparison between the external-IV and internal-moment estimates.	If the two information sources conflict, the overidentifying restrictions can reject but cannot by themselves say which assumption failed.

Option	Main identifying information	What is identified in the scalar triangular model	Main practical cautions
Variance-regime / Rigobon-style identification	A scalar state or instrument Z partitions the sample into variance regimes; the structural impact matrix is fixed while shock variances change nonproportionally.	In a square two-shock representation, the impact ratio for the labeled ω_2 shock can be identified from regime covariance matrices; in the one-changing-shock case it is a simple covariance-difference ratio.	This is an alternative square-shock model, not the three-shock common-factor model. Needs a credible regime/state and shock labeling.
Parametric volatility and unconditional volatility moments	GARCH, Markov-switching, smooth transition, stochastic-volatility, or autocovariances of squared residuals provide additional covariance equations.	Identifies a square impact matrix and hence an impact ratio after normalization and labeling.	Strong functional-form or time-series assumptions; high-moment estimates can be noisy.
Non-Gaussian ICA / likelihood / higher-moment GMM	Independent or partly independent non-Gaussian shocks in a square decomposition of (ω_2, ω_1) .	Identifies the square mixing matrix up to sign, scale, and column order; θ is an impact ratio for the correctly labeled column.	Needs shock labeling, sign normalization, and enough departure from Gaussianity. Standard Wald intervals are unreliable under weak non-Gaussianity.

The bottom line is that there are two direct methods for the exact Lewbel triangular regression problem. Lewbel’s method uses the scalar instrument Z to construct a heteroskedasticity-based instrument. Lee–Lewbel–Schennach–Zhang instead do not need Z for their internal point or set identification; they use a common-factor decomposition of the errors. The methods surveyed by Lewis are valuable alternatives and diagnostics, but they become direct estimators of θ only after one changes the latent structure to a square two-shock representation and labels one shock as the shock whose effect on ω_1 per unit of ω_2 is the desired coefficient.

2 Target scalar triangular system and notation

The notation follows the supplied TeX note, specialized to one endogenous regressor. Let $Y_{1,t+1}$ be the outcome and let $Y_{2,t+1}$ be a scalar endogenous regressor. Let X_t be the common vector of controls used in both the outcome equation and the reduced form for the endogenous regressor. The

triangular model is

$$Y_{1,t+1} = X_t^\top \beta_1 + \theta Y_{2,t+1} + \varepsilon_{1,t+1}, \quad (1)$$

$$Y_{2,t+1} = X_t^\top \beta_2^R + \varepsilon_{2,t+1}. \quad (2)$$

The coefficient of interest is the scalar θ . The errors may be correlated, so ordinary least squares in (1) is not generally causal.

Project both observed variables on the same X_t and define the residuals

$$\omega_{1,t+1} := Y_{1,t+1} - X_t^\top \beta_1^R, \quad \beta_1^R := (\mathbb{E}[X_t X_t^\top])^{-1} \mathbb{E}[X_t Y_{1,t+1}], \quad (3)$$

$$\omega_{2,t+1} := Y_{2,t+1} - X_t^\top \beta_2^R, \quad \beta_2^R := (\mathbb{E}[X_t X_t^\top])^{-1} \mathbb{E}[X_t Y_{2,t+1}]. \quad (4)$$

Under $\mathbb{E}[X_t \varepsilon_{1,t+1}] = \mathbb{E}[X_t \varepsilon_{2,t+1}] = 0$, the residualized system is

$$\omega_{1,t+1} = \theta \omega_{2,t+1} + \varepsilon_{1,t+1}, \quad \omega_{2,t+1} = \varepsilon_{2,t+1}. \quad (5)$$

For any candidate value w of θ , the implied structural residual is

$$\varepsilon_{1,t+1}(w) := \omega_{1,t+1} - w \omega_{2,t+1}. \quad (6)$$

Once θ is known or restricted to a set Θ , the non-endogenous coefficients in the outcome equation are recovered by the same linear map used in the supplied note:

$$\beta_1(w) = \beta_1^R - \beta_2^R w. \quad (7)$$

Thus each candidate w for θ implies one and only one candidate coefficient vector $\beta_1(w)$. This is why a finite interval for θ is the central object in the scalar problem.

Throughout, Z_{t+1} denotes a single observed scalar that may be used as an external instrument, a heteroskedasticity-generating variable, a variance-regime indicator, or a shock-labeling variable depending on the identification strategy. The role of Z_{t+1} is strategy-specific; it is not automatically a valid instrument simply because it is observed.

3 Baseline: Lewbel's scalar heteroskedasticity method

Although the two new attached papers are the main focus, the Lewbel baseline is useful because it fixes the notation and clarifies what it means to have one endogenous regressor and one instrument.

3.1 Exact validity gives a point estimate

Lewbel's exact triangular condition is

$$\text{Cov}(Z_{t+1}, \varepsilon_{1,t+1} \varepsilon_{2,t+1}) = 0, \quad \text{Cov}(Z_{t+1}, \varepsilon_{2,t+1}^2) \neq 0. \quad (8)$$

The first condition says that Z_{t+1} is not correlated with the product of the two structural errors. The second condition says that Z_{t+1} predicts the variance of the first-stage error. Therefore

$$H_{t+1} := (Z_{t+1} - \mathbb{E}Z_{t+1}) \omega_{2,t+1} \quad (9)$$

is a valid constructed instrument for $\omega_{2,t+1}$ in the residualized equation. Indeed,

$$\mathbb{E}[H_{t+1} \varepsilon_{1,t+1}] = \text{Cov}(Z_{t+1}, \varepsilon_{1,t+1} \varepsilon_{2,t+1}) = 0, \quad (10)$$

$$\mathbb{E}[H_{t+1} \omega_{2,t+1}] = \text{Cov}(Z_{t+1}, \varepsilon_{2,t+1}^2) \neq 0. \quad (11)$$

In the scalar residualized system this yields the closed-form population coefficient

$$\theta = \frac{\text{Cov}(Z_{t+1}, \omega_{1,t+1}\omega_{2,t+1})}{\text{Cov}(Z_{t+1}, \omega_{2,t+1}^2)}. \quad (12)$$

With controls, the same calculation is applied after residualizing both $Y_{1,t+1}$ and $Y_{2,t+1}$ on X_t , and the remaining coefficients follow from (7).

3.2 Bounded invalidity gives a scalar interval

The supplied TeX note generalizes the exact condition by allowing a bounded amount of contamination. In the scalar case, assume that for a known $\tau \in [0, 1)$,

$$|\text{Corr}(Z_{t+1}, \varepsilon_{1,t+1}\varepsilon_{2,t+1})| \leq \tau \left| \text{Corr}(Z_{t+1}, \varepsilon_{2,t+1}^2) \right|. \quad (13)$$

Define the observable population quantities

$$L := \text{Cov}(Z_{t+1}, \omega_{1,t+1}\omega_{2,t+1}), \quad Q := \text{Cov}(Z_{t+1}, \omega_{2,t+1}^2), \quad (14)$$

$$S_0 := \text{Var}(\omega_{1,t+1}\omega_{2,t+1}), \quad S_1 := \text{Cov}(\omega_{2,t+1}^2, \omega_{1,t+1}\omega_{2,t+1}), \quad S_2 := \text{Var}(\omega_{2,t+1}^2). \quad (15)$$

For a candidate w , the relative-correlation bound becomes the quadratic inequality

$$(L - Qw)^2 - d\{S_0 - 2S_1w + S_2w^2\} \leq 0, \quad d := \tau^2 \frac{Q^2}{S_2}. \quad (16)$$

If $Q \neq 0$ and $\tau < 1$, the leading coefficient is $Q^2 - dS_2 = Q^2(1 - \tau^2) > 0$, so the set is a closed finite interval whenever the discriminant is nonnegative. Writing

$$a := Q^2 - dS_2, \quad b := -2LQ + 2dS_1, \quad c := L^2 - dS_0, \quad (17)$$

the interval is

$$\Theta_{\text{Lewbel}}(\tau) = \left[\frac{-b - \sqrt{b^2 - 4ac}}{2a}, \frac{-b + \sqrt{b^2 - 4ac}}{2a} \right]. \quad (18)$$

At $\tau = 0$, the interval collapses to the point in (12). As τ grows, the interval widens. The interpretation is simple: the more invalidity one is willing to tolerate in the constructed instrument, the less precisely θ is identified.

3.3 Sampling uncertainty

The exact Lewbel point estimator can be computed by two-stage least squares using X_t and $(Z_{t+1} - \bar{Z})\hat{\omega}_{2,t+1}$ as instruments. Heteroskedasticity-robust or cluster-robust covariance matrices are standard. For the interval (18), a conservative confidence set can be obtained by replacing the population moments with sample moments and using a bootstrap or a moment-inequality outer construction. The attached Lee et al. paper provides a more explicit closed-form confidence construction for its own second-moment set, which is described in Section 5.

4 Direct option 1 from Lee–Lewbel–Schennach–Zhang: point identification by non-Gaussianity

The `trigmm` paper studies the same triangular shape as (1)–(2), but it does not use a Lewbel-style scalar Z to construct an instrument. Instead, it imposes a latent common-factor structure on the residuals. In the notation of this report,

$$\omega_{2,t+1} = u_{t+1} + v_{t+1}, \quad \omega_{1,t+1} = \theta\omega_{2,t+1} + \beta_u u_{t+1} + r_{t+1}. \quad (19)$$

Equivalently,

$$\omega_{1,t+1} = \alpha u_{t+1} + \theta v_{t+1} + r_{t+1}, \quad \alpha := \theta + \beta_u. \quad (20)$$

Here u is the unobserved common cause that creates endogeneity, v is the idiosyncratic first-stage shock, and r is the idiosyncratic outcome shock. The key assumptions are:

- u_{t+1} , v_{t+1} , and r_{t+1} are mutually independent, and in the implemented command they are also independent of X_t ;
- the sign of $\beta_u = \alpha - \theta$ is specified by the researcher;
- sufficiently high moments exist and the relevant higher cumulants are nondegenerate.

The sign condition matters because replacing u by $-u$ changes the sign of the common-factor loading without changing the observed distribution. Economic theory must therefore decide whether the common factor moves Y_2 and Y_1 in the same direction or in opposite directions.

4.1 Cumulant notation

Let $\Phi_{\omega_2, \omega_1}$ be the log characteristic function of $(\omega_{2,t+1}, \omega_{1,t+1})$, and let Φ_{ω_2} be the log characteristic function of $\omega_{2,t+1}$. Define joint cumulants by

$$\Phi_{\omega_2, \omega_1}^{k,l} := \left. \frac{\partial^{k+l} \Phi_{\omega_2, \omega_1}(\zeta, \xi)}{i^{k+l} \partial \zeta^k \partial \xi^l} \right|_{\zeta=0, \xi=0}, \quad \Phi_{\omega_2}^k := \left. \frac{\partial^k \Phi_{\omega_2}(\zeta)}{i^k \partial \zeta^k} \right|_{\zeta=0}. \quad (21)$$

For ordinary intuition, $\Phi_{\omega_2}^2$ is the variance of ω_2 , $\Phi_{\omega_2}^3$ is its third cumulant, and higher orders measure higher-order departures from Gaussianity. Gaussian variables have all cumulants above order two equal to zero.

Because cumulants of independent sums add, the latent model implies, for any nonnegative integer p ,

$$\Phi_{\omega_2}^{3+p} = \kappa_u^{3+p} + \kappa_v^{3+p}, \quad (22)$$

$$\Phi_{\omega_2, \omega_1}^{2+p,1} = \alpha \kappa_u^{3+p} + \theta \kappa_v^{3+p}, \quad (23)$$

$$\Phi_{\omega_2, \omega_1}^{1+p,2} = \alpha^2 \kappa_u^{3+p} + \theta^2 \kappa_v^{3+p}, \quad (24)$$

where κ_u^m and κ_v^m are the order- m cumulants of u and v . Eliminating these nuisance cumulants gives the observable moment equation

$$M_p(\alpha, \theta) := \Phi_{\omega_2, \omega_1}^{1+p,2} - \alpha^2 \Phi_{\omega_2}^{3+p} - (\theta + \alpha) \left(\Phi_{\omega_2, \omega_1}^{2+p,1} - \alpha \Phi_{\omega_2}^{3+p} \right) = 0. \quad (25)$$

This is the scalar version of the moment condition implemented by `trigmm`.

4.2 Identification from two cumulant equations

Two different values of p give two equations in the two unknowns (α, θ) . For example,

$$M_0(\alpha, \theta) = 0, \quad M_1(\alpha, \theta) = 0 \quad (26)$$

can point identify (α, θ) . The default `trigmm` specification uses $p = 0, 1$. The command can also use $p = 0, 1, 2$, which gives three moment equations for two parameters and therefore overidentifying restrictions.

For a pair $q < \tilde{q}$, the essential nondegeneracy condition is that the higher cumulants are not lined up in a way that makes the two equations carry the same information. In the notation here, a convenient statement is

$$\Phi_{\omega_2}^{3+\tilde{q}} \Phi_{\omega_2, \omega_1}^{2+q, 1} \neq \Phi_{\omega_2}^{3+q} \Phi_{\omega_2, \omega_1}^{2+\tilde{q}, 1}. \quad (27)$$

For the default pair $q = 0, \tilde{q} = 1$, this condition can fail if either u or v is Gaussian, if both are symmetric, or if they have exactly the same distribution. This is why the method is described as identification by non-Gaussianity: information must be present beyond means and variances.

Once (α, θ) is identified, the loading of the common factor is

$$\beta_u = \alpha - \theta. \quad (28)$$

The latent variances can be recovered from second moments. Let $\sigma_u^2 := \text{Var}(u)$, $\sigma_v^2 := \text{Var}(v)$, and $\sigma_r^2 := \text{Var}(r)$. Then

$$\sigma_u^2 = \frac{\text{Cov}(\omega_1, \omega_2) - \theta \text{Var}(\omega_2)}{\alpha - \theta}, \quad (29)$$

$$\sigma_v^2 = \text{Var}(\omega_2) - \sigma_u^2, \quad (30)$$

$$\sigma_r^2 = \text{Var}(\omega_1) - \alpha^2 \sigma_u^2 - \theta^2 \sigma_v^2. \quad (31)$$

The nonnegativity and plausible sizes of these implied variances are useful model diagnostics.

4.3 How one scalar external instrument can enter

The internal `trigmm` moments do not require an external instrument. If a scalar external instrument Z_{t+1} is available and satisfies the conventional exclusion condition

$$\mathbb{E}[Z_{t+1}\{\omega_{1,t+1} - \theta\omega_{2,t+1}\}] = 0, \quad (32)$$

then it can be added as an additional GMM moment. With one external instrument and the internal moment pair $M_0 = M_1 = 0$, the model is overidentified. This is a particularly useful implementation because it lets the researcher compare three sources of information:

1. the conventional IV estimate from (32);
2. the internal higher-cumulant estimate from (25);
3. the combined GMM estimate using both.

Agreement across these estimates is not proof, but it is a strong robustness check because the identifying assumptions are very different.

4.4 Inference and computation

The point-identified `trigmm` estimator is a nonlinear GMM estimator. Under strong identification and valid moments, standard GMM covariance formulas give standard errors for θ , α , β_u , the variance components, and the recovered X_t coefficients. The paper emphasizes three practical points.

- The objective function can have multiple local minima. Starting values matter, especially with the overidentified $p = 0, 1, 2$ specification.
- The overidentified specification allows a Hansen J test. When the extra moments come from higher cumulants, a rejection indicates a failure of the latent distributional assumptions or the linear model, not merely a bad external instrument.
- Weak or nearly degenerate higher-moment identification shows up as large standard errors and poorly conditioned moment derivatives. The paper suggests checking singular values of the GMM derivative matrix as a diagnostic.

5 Direct option 2 from Lee–Lewbel–Schennach–Zhang: second-moment set identification

The same common-factor model also gives a set-identified result that uses only second moments. This option is useful when the non-Gaussianity needed for Section 4 is not credible or is too weak to estimate precisely.

The underlying Lee–Lewbel–Schennach–Zhang theory contains a sharp identified set based on decomposing the observed distribution into independent Gaussian and non-Gaussian components. That sharp set is conceptually the most complete second-moment object, but it is difficult to estimate and explain in applied work. The Stata paper therefore focuses on a coarser but transparent outer bound based only on the covariance matrix of (ω_2, ω_1) . The formulas below describe that implemented bound.

Assume first that $\beta_u = \alpha - \theta > 0$. Define

$$B_0 := \frac{\mathbb{E}[\omega_{1,t+1}\omega_{2,t+1}]}{\mathbb{E}[\omega_{2,t+1}^2]}, \quad D_0 := \frac{\mathbb{E}[\omega_{1,t+1}^2]\mathbb{E}[\omega_{2,t+1}^2] - \{\mathbb{E}[\omega_{1,t+1}\omega_{2,t+1}]\}^2}{\{\mathbb{E}[\omega_{2,t+1}^2]\}^2}. \quad (33)$$

If the residuals are mean zero, B_0 is simply the population OLS slope from regressing ω_1 on ω_2 , and D_0 is the residual variance from that regression scaled by $\text{Var}(\omega_2)$.

Because

$$B_0 = \frac{\alpha\mathbb{E}[u^2] + \theta\mathbb{E}[v^2]}{\mathbb{E}[u^2] + \mathbb{E}[v^2]} = \alpha\lambda + \theta(1 - \lambda), \quad \lambda := \frac{\mathbb{E}[u^2]}{\mathbb{E}[u^2] + \mathbb{E}[v^2]} \in [0, 1], \quad (34)$$

we obtain the first bound,

$$\theta \leq B_0 \leq \alpha. \quad (35)$$

The sharper second-moment bound reported in the paper is

$$(B_0 - \theta)(\alpha - B_0) \leq D_0. \quad (36)$$

Thus the population identified set is

$$\Theta_{\text{LSZ,set}}^+ := \{(\alpha, \theta) : \theta \leq B_0 \leq \alpha, (B_0 - \theta)(\alpha - B_0) \leq D_0\}. \quad (37)$$

The plus sign indicates $\beta_u > 0$.

If instead $\beta_u < 0$, apply the positive-loading result to $-\omega_2$. The order reverses and the set becomes

$$\Theta_{\text{LSZ,set}}^- := \{(\alpha, \theta) : \alpha \leq B_0 \leq \theta, (\theta - B_0)(B_0 - \alpha) \leq D_0\}. \quad (38)$$

5.1 Important consequence: the marginal set for θ is usually not a finite interval

The joint set (37) is informative, but its projection onto θ alone is

$$\{\theta : \exists \alpha \text{ such that } (\alpha, \theta) \in \Theta_{\text{LSZ,set}}^+\} = (-\infty, B_0] \quad (39)$$

in population. For $\beta_u < 0$, the projection is $[B_0, \infty)$. Therefore the second-moment set is a joint bound on the causal effect θ and the total common-factor effect α , not a finite scalar interval for θ by itself. A finite interval for θ requires one more substantive restriction, such as a known range for $\beta_u = \alpha - \theta$ or a known range for α .

For example, if $\beta_u > 0$ and $0 \leq \beta_u \leq \bar{\beta}$, then $\alpha = \theta + \beta_u$ adds

$$0 \leq \alpha - \theta \leq \bar{\beta}. \quad (40)$$

Projecting the intersection of (37) and (40) gives a finite interval whenever $\bar{\beta}$ is finite. This extra magnitude restriction is not part of the base `trigmmset` second-moment result; it must come from outside the paper.

5.2 Confidence region used by `trigmmset`

In sample, estimate B_0 and D_0 by plug-in estimators $(\hat{B}, \hat{D})$ and estimate their covariance matrix $\hat{V}$ by the delta method. For a significance level δ , construct the ellipse

$$\mathcal{E}_\delta := \left\{ (B, D) : \begin{bmatrix} B - \hat{B} & D - \hat{D} \end{bmatrix} \hat{V}^{-1} \begin{bmatrix} B - \hat{B} \\ D - \hat{D} \end{bmatrix} \leq \chi_{2,1-\delta}^2 \right\}. \quad (41)$$

Then $\mathcal{E}_\delta$ contains the true (B_0, D_0) with approximate probability $1 - \delta$. The command reports a worst-case region that contains the union of all identified sets generated by all $(B, D) \in \mathcal{E}_\delta$. For $\beta_u > 0$, define

$$B^- := \min_{(B,D) \in \mathcal{E}_\delta} B, \quad B^+ := \max_{(B,D) \in \mathcal{E}_\delta} B, \quad (42)$$

$$D^* := \max_{(B,D) \in \mathcal{E}_\delta} \{D + (B - B^+)(B^- - B)\}. \quad (43)$$

The reported confidence set is the union of four simple regions:

$$\alpha \geq \theta, \quad B^- \leq \alpha \leq B^+, \quad B^- \leq \theta \leq B^+, \quad (44)$$

$$B^- \leq \alpha \leq B^+, \quad \theta \leq B^-, \quad (45)$$

$$B^- \leq \theta \leq B^+, \quad B^+ \leq \alpha, \quad (46)$$

$$D^* \geq (\alpha - B^+)(B^- - \theta), \quad B^+ \leq \alpha, \quad \theta \leq B^-. \quad (47)$$

This construction is conservative but easy to draw and easy to report. It gives a confidence region for the joint object (α, θ) . The marginal projection onto θ is again generally one-sided: approximately $(-\infty, B^+]$ for $\beta_u > 0$ and $[B^-, \infty)$ for $\beta_u < 0$.

5.3 Coefficients on controls

The endogenous-equation slope β_2^R is point identified by the projection of Y_2 on X_t . The coefficient vector on X_t in the outcome equation is

$$\beta_1(\theta) = \beta_1^R - \beta_2^R \theta. \quad (48)$$

Because θ is set identified, β_1 is also set identified. The `trigmmset` paper describes a conservative element-by-element way to bound these coefficients using confidence intervals for the components of β_2^R , β_1^R , and B_0 . In the scalar notation here, if the sign of a particular component of β_2^R is known, the one-sided bound on θ translates into a one-sided bound on that element of β_1 . If the sign of that component is not statistically determined, the corresponding coefficient bound may be uninformative.

6 Options from Lewis's review, translated to the scalar triangular setting

Lewis's Annual Review paper is written for structural vector autoregressions and simultaneous systems with a square impact matrix. It is therefore not identical to the Lewbel triangular model in (1)–(2), and it is also not identical to the Lee–Lewbel–Schennach–Zhang three-shock common-factor model. To use those options in the present scalar problem, impose an alternative square decomposition of the two residuals:

$$U_{t+1} := \begin{bmatrix} \omega_{2,t+1} \\ \omega_{1,t+1} \end{bmatrix} = B\eta_{t+1}, \quad B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}. \quad (49)$$

The shocks $\eta_{t+1} = (\eta_{1,t+1}, \eta_{2,t+1})^\top$ are normalized in some way, such as unit variances or a unit diagonal. After estimating B , the scalar coefficient analogous to θ is an impact ratio for a labeled shock. If column j is interpreted as the shock that moves ω_2 , then

$$\theta_j := \frac{b_{2j}}{b_{1j}}, \quad b_{1j} \neq 0. \quad (50)$$

This ratio is not automatically the structural θ in (5); it becomes the target only if the researcher is willing to interpret the selected shock as the exogenous variation in ω_2 whose causal impact on ω_1 is sought. Shock labeling is therefore central.

6.1 Heteroskedasticity option A: two variance regimes

Let the scalar variable Z_{t+1} partition the data into two regimes, say $Z_{t+1} = 0$ and $Z_{t+1} = 1$. Suppose

$$\Sigma_r := \text{Var}(U_{t+1} \mid Z_{t+1} = r) = B\Lambda_r B^\top, \quad \Lambda_r = \text{diag}(\lambda_{r1}, \lambda_{r2}), \quad r = 0, 1. \quad (51)$$

The impact matrix B is constant across regimes, but the shock variances change. The two-regime heteroskedasticity condition is

$$\frac{\lambda_{11}}{\lambda_{01}} \neq \frac{\lambda_{12}}{\lambda_{02}}. \quad (52)$$

This says that the two shock variances do not change proportionally across the two regimes. Then

$$\Sigma_1 \Sigma_0^{-1} = B\Lambda_1 \Lambda_0^{-1} B^{-1}, \quad (53)$$

so the columns of B are eigenvectors of an observable matrix, up to scale and order. Once a column is labeled as the ω_2 shock, $\theta_j = b_{2j}/b_{1j}$ is identified.

A particularly transparent special case is that only the variance of the labeled shock j changes. Then

$$\Sigma_1 - \Sigma_0 = (\lambda_{1j} - \lambda_{0j})b_j b_j^\top, \quad (54)$$

where b_j is the j -th column of B . With $U = (\omega_2, \omega_1)^\top$, the impact ratio is

$$\theta_j = \frac{\text{Cov}(\omega_2, \omega_1 \mid Z = 1) - \text{Cov}(\omega_2, \omega_1 \mid Z = 0)}{\text{Var}(\omega_2 \mid Z = 1) - \text{Var}(\omega_2 \mid Z = 0)}. \quad (55)$$

This is the closest variance-regime analogue of a single-instrument estimator. The scalar Z is not an excluded IV; it supplies a high-variance and a low-variance state.

6.2 Heteroskedasticity option B: more regimes, Markov switching, smooth transition, and GARCH

The same covariance-equation logic extends in several directions.

- **More than two regimes.** Additional regimes add more covariance matrices. If B is constant and the diagonal variance changes are rich enough, B is overidentified. Overidentifying restrictions can test whether the fixed impact matrix and diagonal shock-variance assumptions are plausible.
- **Markov-switching variances.** The regimes are latent rather than observed. Identification still comes from distinct variance matrices satisfying a nonproportionality condition, but estimation must also infer the regimes.
- **Smooth transition variances.** The covariance matrix evolves as

$$\Omega_t = (1 - F(s_t))\Sigma_0 + F(s_t)\Sigma_1, \quad (56)$$

where s_t is a transition variable. Conditional on the transition parameters, the endpoints Σ_0 and Σ_1 identify B by the same logic as Section 6.1. Identification is weak if the transition is nearly flat or if s_t does not track actual volatility changes.

- **GARCH or stochastic volatility.** Parametric volatility models use the full time path of conditional variances. The maintained idea remains fixed B plus diagonal, time-varying structural variances. Parametric assumptions can improve efficiency but introduce functional-form risk.

In all of these variants, one can recover an impact ratio such as (50) only after normalizing and labeling the relevant column of B .

6.3 Heteroskedasticity option C: unconditional moments of squared residuals

Lewis also reviews distribution-free identification based on persistence in volatility. In the present two-variable notation, the useful objects are autocovariances of squared and cross-product residuals, such as

$$\text{Cov}\left(\text{vec}(U_{t+1}U_{t+1}^\top), \text{vec}(U_{t+1-s}U_{t+1-s}^\top)\right), \quad s > 0. \quad (57)$$

If structural shocks themselves are serially uncorrelated but their variances are persistent, these autocovariances reveal how the structural variances move over time. Identification requires the volatility processes of the shocks not to have proportional persistence patterns. This option is attractive because it does not require choosing exact regimes, but it is demanding in finite samples because it uses fourth-order quantities and their autocovariances.

6.4 Non-Gaussianity option A: independent component analysis

Instead of changing variances over time, assume that the components of η_{t+1} in (49) are mutually independent and that at most one of them is Gaussian. Then, by the Darmois–Skitovich/Comon logic reviewed by Lewis, the decomposition $U = B\eta$ is unique up to scale, sign, and column order. In the two-shock case this means that at least one shock must be non-Gaussian.

Independent component analysis (ICA) estimates the rotation of the whitened residual vector that makes the candidate shocks as independent or as non-Gaussian as possible. After estimating B , the scalar causal parameter is again an impact ratio b_{2j}/b_{1j} for the economically labeled column. The single scalar Z can help label the column, for example by choosing the shock most correlated with Z or the shock whose response signs match prior theory, but the statistical identification itself comes from independence and non-Gaussianity, not from the exclusion restriction $\mathbb{E}[Z\varepsilon_1] = 0$.

6.5 Non-Gaussianity option B: likelihood and pseudo-likelihood

A parametric likelihood version specifies densities for the independent shocks, for example Student- t densities. If the density is correctly specified, maximum likelihood can be efficient. Pseudo-maximum likelihood can remain consistent under certain misspecifications, provided the pseudo-model is itself identified. The cost is that there are many possible ways to be non-Gaussian, so a particular density choice is an additional modeling decision.

In the scalar triangular report, this option again delivers θ only through an estimated square mixing matrix and a labeled impact ratio. It is not the same as the Lee–Lewbel–Schennach–Zhang cumulant equations, which are tailored to the three-component common-factor model.

6.6 Non-Gaussianity option C: moment-based GMM with coskewness or cokurtosis

Moment-based non-Gaussian identification selects particular independence-implied moments. Let $\eta = CU$ denote candidate shocks after applying an unmixing matrix $C = B^{-1}$ and normalizing variances to one. Typical restrictions include

$$\mathbb{E}[\eta_1\eta_2] = 0, \tag{58}$$

$$\mathbb{E}[\eta_1^2\eta_2] = 0, \quad \mathbb{E}[\eta_1\eta_2^2] = 0, \tag{59}$$

$$\mathbb{E}[\eta_1^2\eta_2^2] - 1 = 0, \quad \mathbb{E}[\eta_1^3\eta_2] = 0, \quad \mathbb{E}[\eta_1\eta_2^3] = 0. \tag{60}$$

The restrictions in (59) use skewness-type information; those in (60) use kurtosis-type information. Identification requires at least one shock to have relevant nonzero skewness or nonzero excess kurtosis. A GMM estimator chooses C so these cross-moments are close to zero, then obtains $B = C^{-1}$ and the impact ratio (50).

This option is transparent because the identifying moments are explicit, but it can be noisy: estimating fourth moments and the covariance matrix of moment conditions may require moments up to order eight.

6.7 Non-independent components and coheteroskedasticity

The Annual Review also discusses newer work relaxing full independence. The core idea is to replace full independence by restrictions on higher-order cumulant tensors, such as diagonal cumulant tensors or reflectional-invariance patterns. These methods are promising because they can allow some forms of dependence, including coheteroskedasticity. In the scalar triangular problem they

are best viewed as advanced alternatives to the simple square ICA model, not as direct Lewbel or `trigmm` estimators.

7 Using one scalar instrument with each option

The same observed scalar Z can play different roles. Confusing these roles is the easiest way to misuse these methods.

Role 1. External IV. If $\mathbb{E}[Z\varepsilon_1] = 0$ and $\mathbb{E}[Z\omega_2] \neq 0$, then Z directly identifies θ by conventional IV:

$$\theta_{\text{IV}} = \frac{\mathbb{E}[Z\omega_1]}{\mathbb{E}[Z\omega_2]}. \quad (61)$$

This moment can be combined with `trigmm`'s internal moments.

Role 2. Lewbel heteroskedasticity shifter. If Z satisfies (8), then $(Z - \mathbb{E}Z)\omega_2$ is the constructed instrument. Here the relevant covariance is $\text{Cov}(Z, \omega_2^2)$, not $\mathbb{E}[Z\omega_2]$.

Role 3. Variance-regime indicator or volatility state. If Z defines high- and low-volatility regimes, then it supplies two covariance matrices. The assumption is not exclusion but fixed B and nonproportional changes in shock variances.

Role 4. Shock-labeling variable. In ICA or non-Gaussian square models, Z may help decide which column of B is economically meaningful. It does not by itself identify the rotation unless an additional IV or moment condition is imposed.

Role 5. No role in internal LSZ identification. The Lee–Lewbel–Schennach–Zhang point and set identification results do not require Z . They can be used when no external instrument exists. A valid Z can be added, but it is optional.

8 Inference and weak-identification issues

All the methods in this report use information beyond the first two unconditional moments of a homoskedastic linear regression. That information can be weak.

Lewbel exact and bounded-invalidity inference. If $\text{Cov}(Z, \omega_2^2)$ is near zero, the constructed instrument is weak. Standard IV weak-instrument diagnostics are informative because the estimator is implemented as two-stage least squares with the constructed instrument. In the bounded invalidity interval, the same weak relevance makes the interval wide and can make sample roots unstable.

Lee–Lewbel–Schennach–Zhang point inference. If the relevant higher cumulants of u and v are small or nearly degenerate, the GMM objective is flat in some direction. Then standard errors can be large, local minima become a practical problem, and Wald intervals can be misleading. The overidentified $p = 0, 1, 2$ specification, the Hansen J test, and singular values of the moment derivative matrix are useful diagnostics, but none proves identification strength in finite samples.

Lee–Lewbel–Schennach–Zhang set inference. The `trigmmset` confidence region is explicitly set-valued. It first builds an ellipse for (B_0, D_0) and then reports a conservative worst-case joint region for (α, θ) . This is more honest than reporting a spurious finite scalar confidence interval for θ when the population identified set itself projects to a half-line.

Variance-regime inference. If variances change little, or if both shock variances change almost proportionally, the eigenvectors in (53) are weakly identified. Robust GMM confidence sets or weak-identification-robust procedures are preferable to simple Wald intervals.

Non-Gaussian inference. If the shocks are close to Gaussian, independence-based rotations are weakly identified. Testing for non-Gaussianity and independence is useful but difficult because the shocks are estimated objects. Robust score/projection methods have been developed in the literature, but they are more complex than ordinary standard errors.

9 Software and implementation notes

- **Stata.** Lee et al. provide `trigmm` for the point-identified non-Gaussian common-factor method and `trigmmset` for the second-moment set. The commands are wrappers around Stata’s `gmm` infrastructure and use Stata post-estimation tools. Lewbel’s heteroskedasticity-generated instruments are implemented in `ivreg2h`.
- **R or Python.** The scalar formulas in Sections 3 through 5 are simple enough to implement directly. For the LSZ point estimator, one needs routines for sample cumulants and nonlinear GMM. For set estimation, one needs only plug-in estimates of B_0 and D_0 , their delta-method covariance matrix, and a routine to draw or optimize over the reported joint region. Generic GMM, nonlinear optimization, and bootstrap tools are sufficient, but a direct off-the-shelf implementation of `trigmm` for R is not part of the attached papers.
- **ICA and SVAR methods.** Generic ICA, VAR, GARCH, and structural-VAR software can estimate some of the broader Lewis-type models. The researcher must still map the estimated impact matrix into the scalar ratio (50) and justify the shock label. These packages do not automatically estimate the Lewbel triangular coefficient θ .

10 A practical decision tree

1. **Start with the maintained triangular model.** Residualize both $Y_{1,t+1}$ and $Y_{2,t+1}$ on the same X_t and work with $\omega_{1,t+1}, \omega_{2,t+1}$.
2. **If a conventional scalar IV is credible, use it and report it.** Estimate (61) or its controlled version. Then consider adding Lewbel or `trigmm` moments as robustness checks or efficiency-improving overidentifying restrictions.
3. **If no conventional IV is credible but a heteroskedasticity shifter is credible, use Lewbel.** Check whether $\text{Cov}(Z, \omega_2^2)$ is meaningfully nonzero. If exact validity is doubtful, report the bounded-invalidity interval over a grid of τ values.
4. **If the common-factor story is credible and the errors are plausibly non-Gaussian, use trigmm.** Specify the sign of β_u , estimate with $p = 0, 1$ and with $p = 0, 1, 2$, vary starting values, and report diagnostics.
5. **If the common-factor story is credible but non-Gaussianity is not, use trigmmset.** Report the joint region for (α, θ) and state clearly whether the implied marginal set for θ is one-sided. Do not report a finite interval for θ unless an additional magnitude restriction is imposed.

6. **If the data are time series with credible volatility regimes or persistent volatility, consider the Lewis heteroskedasticity options.** Make clear that this replaces the common-factor model by a square impact-matrix model and that θ is an impact ratio for a labeled shock.
7. **If non-Gaussian independent shocks are credible, consider ICA, likelihood, or higher-moment GMM.** Again, report how the shock was labeled and how the impact ratio was normalized.

11 Conclusion

In the scalar triangular setting, the methods differ mainly in what replaces the missing excluded instrument. Lewbel uses a scalar heteroskedasticity shifter to construct an instrument. Lee–Lewbel–Schennach–Zhang use a common-factor error structure: higher cumulants yield point identification, while second moments yield a conservative joint set. Lewis’s review offers a menu of broader higher-moment strategies—variance regimes, volatility processes, ICA, likelihood, and moment-based non-Gaussian GMM—but those strategies require a square shock decomposition and a shock-labeling step before they deliver a scalar effect comparable to θ .

The most transparent reporting strategy is to state, for each estimate or set, which object identifies θ : an exclusion moment, a heteroskedasticity covariance, a non-Gaussian cumulant system, a second-moment common-factor inequality, a variance-regime covariance change, or an independent-component rotation. Once that distinction is clear, all other coefficients follow from the simple recovery formula $\beta_1(w) = \beta_1^R - \beta_2^R w$.

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